

Abstract: In Kere (Trans-New Guinea, Papua-New Guinea) three phrasal phonological processes interact in an opaque counterfeeding pattern. This is unexpected under many approaches to phonology, which predict the absence of strictly phrase-level opacity. In this squib, I show that this pattern is expected if gradient symbolic representations are assumed and spreading leads to loss of gradient activity on tone-bearing units, which share the activity of the linked tone among themselves. This reduces markedness constraint violations and causes underapplication of interacting processes. The account predicts phrase-level opacity to always involve a process modeled as spreading.

Keywords: phonology, opacity, tone, gradience, Papuan languages, phrasal phonology

1 Phrase-level Opacity and Phonological Theory

A phonological process is opaque (roughly) if it either fails to apply even though its context is given on the surface or it applies even though its context is not present on the surface (cf. e.g. Kiparsky, 1973; Baković, 2007; Baković & Blumenfeld, 2022). This was modeled in rule-based theory as extrinsic rule ordering. Certain patterns of phonological opacity pose a challenge to almost all theories of phonology that do not include extrinsic ordering of phonological rules, such as Standard Parallel Optimality Theory (Prince & Smolensky, 1993; McCarthy & Prince, 1993, 1995).¹ This is even more pressing for phrase-level patterns of opacity, since — because most versions do away with extrinsic rule ordering — they only predict a limited amount of opacity by introducing designated devices. Strictly phrase-level opacity is usually excluded. These devices are morphologically-defined strata in order to capture phonological opacity across these domain boundaries for Stratal OT and Lexical Phonology without extrinsic rule ordering at the phrase level (Clark, 1990; Kiparsky,

2015; Bermúdez-Otero, in preparation), but also Base-Derivative-Correspondence for Base-Derivative-Faithfulness (Benua, 1997), which is only defined between words and sub-words. This leads to the No Phrase-Level Opacity Hypothesis (NPLOH) in (1).

(1) **No Phrase-Level Opacity Hypothesis**

A process X and a process Y cannot interact opaquely if both X and Y apply at the phrasal stratum.

In this squib, I will present a counterexample to this hypothesis from phrase-level opacity in Kere tonal phonology, where three processes interact in a counterfeeding pattern, i.e. one process creates the context for another, but the latter process does not apply. I will also show that this pattern is expected under the independently introduced notion of Shared Activity (Goldrick & Smolensky, 2016; Faust & Smolensky, 2017; Zimmermann, 2021) in Gradient Symbolic Representations (Goldrick & Smolensky, 2016; Rosen, 2016). Tonal representations in the present approach include gradient activity that is shared among TBUs associated to the same tone. This predicts the following: if shared activity is the only source of phrase-level opacity, then all phrase-level opacity should involve at least one spreading process. It thus leads to a more restrictive theory of opacity compared to extrinsic rule ordering. In the rest of this squib, I will introduce the data from Kere in section 1, and the theoretical assumption of Shared Activity in section 4; followed by an analysis in section 4, further applications in section 5, and a conclusion in section 6.

2 Phrase-level Opacity in Kere

The phrase level phonology of Kere (Trans-New Guinea, Papua-New Guinea), also known as Sinasina, features a tonal process, so-called Upstep, which causes a global

rise in pitch that affects all following tones. This is indicated by $\uparrow$ before the upstepped TBU in the examples in (2), which give the phrasal input and output. All data and empirical generalizations are from Rarrick (2017) and associated files, henceforth SR. The high tone in the word /íglà/ ‘in’ is upstepped after a preceding high-toned TBU in (2-a).² Similarly, the high tone on /tái/ ‘INDEF.DET’ is upstepped in (2-b).

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|---|--|
| <p>(2) Upstep (SR, 117,118,1-007)</p> <p>a. níl íglà → níl $\uparrow$íglà
 water in
 ‘in the water’
 (cf. yòblágná íglà ‘inside our bones’)</p> <p>b. kàgé tái → kàgé $\uparrow$tái
 times INDEF.DET
 ‘sometimes’
 (cf. págài tái ‘somewhat far’)</p> | <p>(3) Low Deletion (SR 121,120,017-001)</p> <p>a. níl màré → níl mǎré
 water near
 ‘near the water’
 (cf. màré dímiyà ‘to be near’)</p> <p>b. ìgé gòlé → ìgé gǎlé
 house old
 ‘old house’</p> |
|---|--|

Low Deletion³, which changes a single low tone to a high tone if it appears between two high-toned TBUs, interacts opaquely with Upstep. If the high-toned TBU /níl/ ‘water’ occurs before a sequence of a low-toned TBU and a high-toned TBU, the TBU shows up with a high tone, as in (3-a). The same applies to the first TBU of /gòlé/ ‘old’ in (3-b).⁴ In terms of Benua (1997); McCarthy (1998) this constitutes a case of underapplication, since an alternation (i.e. Upstep) does not apply, even though its context specification — two adjacent high tones — is met. In derivational terms, this is a case of counterfeeding opacity because Low Deletion would be expected create the context for (i.e. feed) Upstep.

A third process — Boundary Tone Shift — only occurs at the end of an “utterance, which is a declarative statement” (SR, 148) and inserts a low tone on the final TBU, so that any high tone that was linked to the final TBU associates to the penultimate TBU potentially overwriting previous low tones. Crucially, this process can create the context for either Upstep or Low Tone Deletion but neither applies. In the example

in (4-a), the word /màré/ ‘near’ is assigned a low boundary tone and Boundary Tone Shift creates two adjacent high tones without any Upstep. Boundary Tone Shift and Low Deletion interact in a similar fashion. In (4-b), the final high tone of /òné/ ‘very’ is shifted onto the penultimate TBU, which creates a high-low-high sequence. This demonstrates that Boundary Tone Shift counterfeeds both Upstep and Low Deletion.

- (4) Boundary Tone Shift (SR, 148,124)

a. níl màré% → níl márè%
 water near
 ‘near the water’

b. ìgé kwì òné% → ìgé kwì ónè%
 house new very
 ‘very new house’

(5) Phrase-level tone Opacity

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graph TD
    A[Boundary tone shift] -- counterfeeds --> B[Low deletion]
    B -- counterfeeds --> C[Upstep]
    A -. counterfeeds .-> C

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As shown in (5), Boundary tone shift counterfeeds Low Deletion and Upstep, and Low deletion counterfeeds Upstep.⁵ This highly opaque interaction challenges the NPLOH.⁶

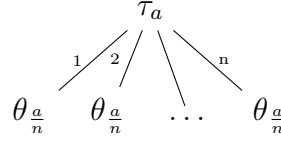
3 GSR and Shared Activity

One possible perspective on this pattern is as a kind of phonologically derived-environment blocking, i.e. a derived pattern is not repaired whereas an underlying pattern is repaired. This can be formalized in Gradient Symbolic Representations (GSR, Goldrick & Smolensky, 2016; Rosen, 2016) as a loss of activity by spreading or shifting. More concretely, I will assume that the the number of TBUs a tone is associated to determines the activity of these TBUs. A definition is given in (6).

(6) **Shared Activity**

The activity a of a TBU θ depends on the number n of TBUs associated to the tone τ that is associated to the TBU θ , i.e. $a(\theta) = \frac{a(\tau)}{n}$.

(7) Shared Activity in Tone spreading



This is based on previous work on reduplication as distributed activity (Zimmermann, 2021) and shared activity inside a morpheme (Goldrick & Smolensky, 2016; Faust & Smolensky, 2017; Rosen, 2021). In Zimmermann (2021), the output activity of a reduplicated form depends on the number of reduplicants: $\frac{1}{2} = 0.5$ in simple reduplication and $\frac{1}{3} = 0.\bar{3}$ in ‘triplication’.⁷ In the present approach, the activity of TBUs is analogously dependent on the number of TBUs associated to the tone.

The line of work exemplified by Goldrick & Smolensky (2016); Faust & Smolensky (2017) and Rosen (2021) similarly assumes that segments in a single morpheme can share activity in the input and this activity is distributed among these segments in the output, such that only some segments surface with full activity and others do not surface at all since no activity is ‘left’. In the present case the activity a of a tone τ is distributed evenly among the TBUs $\theta_1 \dots \theta_n$ associated to it, as illustrated in (7).

Following Zimmermann (2018a,b, 2019, 2021), I view activity as an orthogonal property of phonological objects, which might or might not be interpreted in the phonetic component, similar to other representations, such as syllable structure. In contrast to Zimmermann, I assume that the activation of at least some phonological objects follows predictably from the phonological structure they occur in. For an overview of the more general research program of gradient representations, see Hsu (2022).

A brief note on tonal representations: I will notate three tonal primitives here: H, L, and $^{\uparrow}$ H, associated to a TBU, i.e. a syllable or a mora.⁸ For reasons of space, I will not delve into the specifics of subtonal features (see e.g. Pulleyblank, 1986; Meyase,

2021) and simply assume that high tones and upstepped high tones share some tonal feature [+HIGH] and differ in a register feature [$\pm$ high] (see Yip (1980), Snider (1990), and Lionnet (2022) for similar approaches).

A second crucial assumption concerns the difference between spreading and shifting. I take this to be a minimal one in Kere, namely that shifting involves insertion of a boundary tone at the original tone bearing unit. This means that the effects of spreading and shifting on activity and the interaction with other processes are unified. This allows us to formulate two gradient constraints, *HH and *HLH, which need to refer to TBUs, given in (8). This is clear for the *HLH constraint (8-a) because Low Deletion explicitly targets singly-linked low tones, i.e. low tones associated to a single TBU.⁹ The constraint *HH (8-b) is violated by a sequence of two (non-upstepped) high-toned TBUs and will be repaired by Upstep. In both constraints, violation depend on the activity of the TBUs involved. This is implemented by adding up the activity of all TBUs involved and dividing it by their number. The theory of gradient symbolic representations is usually combined with the framework of Harmonic Grammar (Smolensky & Legendre, 2006). The numbers given to the right of the constraint show its constraint weight. Both constraints have a relatively high weight.

(8) Gradient Constraints

- a. *HLH: Count $\frac{x+y+z}{3}$ violations for a TBU with activity y linked to a low tone following a TBU with activity x and preceding a TBU with activity z both linked to different high ([+HIGH]) tones. 63
- b. *HH: Count $\frac{x+y}{2}$ violations for a TBU with activity x and a following TBU with activity y , linked to different non-upstepped high tones ([+HIGH, -high]) tones. 80

In addition to gradient *HLH and *HH, categorical constraints are needed, as given in (9). NOSPREAD (9-a) and IDENT(register) (9-b) militate against spreading and changes of register features respectively, both with a relatively low weight, with IDENT(r) outweighing NOSPREAD. High-weighted *H% (9-c) enforces low boundary tone assignment, whereas high-weighted NOSPREADR (9-d) and MAX(H) (9-e) control the direction (cf. Zaleska, 2018) and the target of spreading, respectively.

(9) Categorical constraints

- a. NOSPREAD: Count -1 for a high tone associated to more than one TBU. 1
- b. IDENT(register): Count -1 for an output $\uparrow$ ([+high]) that is not in the input. 61
- c. *H%: Count -1 for any phrase-final high tone. 100
- d. NOSPREADR: Count -1 for an output-only association line to the right of another association line. 100
- e. MAX(H): Count -1 for any high tone in the input that is not in the output.

IDENT(r) (cf. Bickmore, 1999; de Lacy, 2002) is the only faithfulness constraint crucial for the current analysis. Nevertheless, the ranking still includes MAX and/or DEP constraints for tone (Myers, 1997).¹⁰ Spreading of tones is also restricted by a CRISPEGE constraint (Itô & Mester, 1999) — not listed above — since tone spreading in Kere is phrase-bound (SR, 132).

Crucially, gradient and categorical constraints are needed (Goldrick & Smolensky, 2016).¹¹ Using only categorical constraints ignores shared activity and — consequently — derivational history fails to derive opaque interactions. Using only gradient constraints would lead to weighting paradoxes similar to the symmetric trade-off effects (cf. Smolensky (2006), Pater (2009, 2016), Murphy (2017), and Müller (2020)).

4 Phrase-level Opacity as Shared Activity

The general idea of Harmonic Grammar is illustrated by the tableau on Upstep in (10). The tableau includes an input I with two high-toned TBUs and two candidates (10-a) and (10-b), where candidate (10-a) is identical to the input but candidate (10-b) introduces an upstep. It also includes constraints with their respective weights.

(10) Upstep

	$\begin{array}{c} H_1 \quad H_1 \quad L_1 \\ \quad \quad \\ I: \quad nil_1 \quad i_1 gla_1 \\ \mathcal{W}= \end{array}$	NO SPREAD R	MAX(H)	*HH	ID(r)	$\mathcal{H}$
		100	100	80	61	
a.	$\begin{array}{c} H_1 \quad H_1 \quad L_1 \\ \quad \quad \\ nil_1 \quad i_1 gla_1 \end{array}$			-1.0		-80.0
b.	$\begin{array}{c} H_1 \quad \uparrow H_1 \quad L_1 \\ \quad \quad \\ nil_1 \quad i_1 gla_1 \end{array}$				-1.0	-61.0
c.	$\begin{array}{c} \quad \quad H_1 \quad L_1 \\ \quad \quad / \quad \backslash \\ nil_{0.5} \quad i_{0.5} gla_1 \end{array}$		-1.0			-100.0
d.	$\begin{array}{c} H_1 \quad H_1 \\ \quad / \quad \backslash \\ nil_1 \quad i_{0.5} gla_{0.5} \end{array}$	-1.0		-0.75		-160.0

Candidate (10-a) has a violation of the higher weighted constraint *HH (indicated by -1) which leads to an overall harmony score of $(-1 \times 80 =) -80$. This is worse than the optimal candidate (10-b) with a score of $(-1 \times 61 =) -61$ that only violates the faithfulness constraint IDENT(r) by introducing an upstep. (10-c) and (10-d) show that rightwards spreading and high tone deletion are not optimal as they incur violations of the high-weighted constraints NOSPREADR and MAX(H), respectively.¹²

Shared Activity is relevant for the counterfeeding interaction between Low Deletion and Upstep in (11). Candidate (11-a) is identical to the input and includes a violation of the constraint *HLH. In order to repair this, candidate (11-b) spreads the second high tone and deletes the intervening low tone. This violates the constraint *HH only gradually. The second and third TBUs share the activity of their high tone now, which means that each of these has an activity of 0.5. Therefore, the violation of the constraint *HH is $-0.75(=\frac{1+0.5}{2})$, which is less severe than for singly-linked tones.

(11) Low Deletion counterfeeds Upstep

$\begin{array}{ccc} H_1 & L_1 & H_1 \\ & & \\ \text{nil}_1 & \text{ma}_1 & \text{re}_1 \end{array}$		MAX(H)	*HH	*HLH	ID(r)	NoSpr	$\mathcal{H}$
$\mathcal{W} =$		100	80	63	61	2	
$\begin{array}{ccc} H_1 & L_1 & H_1 \\ & & \\ \text{nil}_1 & \text{ma}_1 & \text{re}_1 \end{array}$				-1.0			-63.0
$\begin{array}{ccc} H_1 & & H_1 \\ & \nearrow & \\ \text{nil}_1 & \text{ma}_{0.5} & \text{re}_{0.5} \end{array}$			-0.75			-1.0	-62.0
$\begin{array}{ccc} H_1 & & \uparrow H_1 \\ & \nearrow & \\ \text{nil}_1 & \text{ma}_{0.5} & \text{re}_{0.5} \end{array}$					-1.0	-1.0	-63.0
$\begin{array}{ccc} & & H_1 \\ & \nearrow & \\ \text{nil}_{0.3} & \text{ma}_{0.3} & \text{re}_{0.3} \end{array}$		-1.0				-1.0	-102

It is calculated as the activity of the first TBU (=1) plus the activity of the second TBU (=0.5) divided by the number of TBUs associated to said tone (=2). Even with the additional violation of the low weighted NOSPREAD constraint, the overall harmony score of -62 is better than the faithful candidate. This is also a higher harmony score than the upstep candidate (11-c) which has a relatively costly violation

of the categorical constraint IDENT(r). As a reviewer notes, spreading further to the left (11-d) would remedy all markedness violations. Yet it fatally violate the high-weighted constraint MAX(H).

In tableau (12), the faithful candidate (12-a) includes two low-toned TBUs between two high-toned TBUs and fatally violates the boundary tone constraint *H%.¹³ Shifting tones, modeled as spreading in the optimal candidate (12-b), leads to a violation of *HLH albeit only a gradient non-fatal one of $0.8\bar{3}$ ($=\frac{1+1+0.5}{3}$). Candidate (11-c) satisfies the *HLH constraint but yields a slight yet fatal violation of $0.\bar{6}$ ($=\frac{0.\bar{3}+1}{2}$) of the high-weighted *HH constraint. Again, the violation of IDENT(r) excludes Upstep in (12-d). Spreading further in (12-e) again violates high-weighted MAX(H).

(12) Boundary Tone Shift counterfeeds Low Deletion

$\begin{array}{ccccc} L_1 & H_1 & L_1 & L_1 & H_1 \\ & & & & \\ i_1 & g e_1 & k w i_1 & o_1 & n e_1 \end{array}$ $\mathcal{W} =$	MAX(H)	*H%	*HH	*HLH	ID(r)	NoSPR	$\mathcal{H}$
$\begin{array}{ccccc} L_1 & H_1 & L_1 & L_1 & H_1 \\ & & & & \\ i_1 & g e_1 & k w i_1 & o_1 & n e_1 \end{array}$		-1.0					-100.0
$\begin{array}{ccccc} L_1 & H_1 & L_1 & H_1 & L_1 \\ & & & / \backslash & \\ i_1 & g e_1 & k w i_1 & o_{0.5} & n e_{1.5} \end{array}$			$-0.8\bar{3}$			-1.0	-54.5
$\begin{array}{ccccc} L_1 & H_1 & & H_1 & L_1 \\ & & & / \backslash & \\ i_1 & g e_1 & & k w i_{0.3} & o_{0.3} n e_{1.3} \end{array}$			$-0.\bar{6}$			-1.0	$-55.\bar{3}$
$\begin{array}{ccccc} L_1 & H_1 & & \uparrow H_1 & L_1 \\ & & & / \backslash & \\ i_1 & g e_1 & & k w i_{0.3} & o_{0.3} n e_{1.3} \end{array}$					-1.0	-1.0	-63.0
$\begin{array}{ccccc} L_1 & & & H_1 & L_1 \\ & / \backslash & / \backslash & / \backslash & \\ i_1 & g e_{0.25} & k w i_{0.25} & o_{0.25} & n e_{1.25} \end{array}$	-1.0					-1.0	-102.0

I have thus shown that the counterfeeding pattern in Kere phrasal phonology can be derived by assuming Shared Activity in Gradient Symbolic Representations. This

is due to two crucial factors. First, activity allows us to representationally encode minimal information about the derivational history of a phonological object (Smolensky et al., 2014; Trommer, 2019). A TBU that got its tone by spreading has less activity than a TBU that is singly-linked to a tone. Second, gradient violations of markedness constraints allow for a gradient variety of so-called counting effects in tonal phonology (cf. McPherson, 2016). If a candidate violates some lower weighted constraint more severely than some other candidate violates some higher weighted constraint, the former candidate can still become optimal.

5 Further Applications of Shared Activity

The concept of Shared Activity can be applied to similarly opaque phenomena in other languages. As an example outside of tone in phrasal phonology, take other iterative processes, e.g. vowel harmony. Vowel harmony often becomes weaker if a vowel feature is spread out across more vowels, in a pattern of “petering out” (McCollum, 2018, 2019a,b; Kiparsky, 2023), involving phonological weakness and phonetic weakness.

Phonological weakness can involve the loss of iterativity, the emergence of blockers or more restricted locality conditions (Mullin, 2011). This could be modeled if the constraint $\text{MAX}(\text{F})_g$ in (13) prefers output activation to stay on its original host by penalizing the loss of activity, which results from spreading.¹⁴ As a ganging effect with other markedness constraints, this could block spreading only in cases where it would lead to very long spans and a violation of an otherwise irrelevant blocking constraint.¹⁵

- (13) $\text{MAX}(\text{F})_g$: Assign $x - y$ violation for every output feature bearing unit F with output activity y and input activity x , where F is connected to the same feature G in the input and in the output.

Phonetic weakness can involve reduction in an acoustic dimension that is affected by vowel harmony. McCollum (2019a,b) finds such a pattern for the second formant in Uyghur backness harmony.¹⁶ A minimal modification of the present account could predict similar patterns by allowing an uneven distribution of activity between vowels associated to the same back feature. If $\text{MAX}(\text{F})_g$ favors activation on the original host of the feature, spreading will lead to less activation on vowels that were spread to. If it is counteracted by a constraint such as **EVEN!** in (14), which scales with the ratio between gradient activations of segments associated to a given segment, this could explain why only some languages show an uneven distribution with effects in the phonetics (McCollum, 2019a), depending on their relative weights. Interpretation of activity as the strength of the articulatory target would still need to be relative to the activity of preceding targets in order to exclude patterns with even distribution leading to general reduction of the acoustic target in a spread out feature span.

- (14) **EVEN!**: Assign $\frac{x}{y}$ activation for a pair of adjacent segments $\{S_x S_y\}$ connected to the same feature, where $x \geq y$.

Even though the present account lends itself to applications in different subfields of phonology, it still predicts restrictions on possible phrase-level opacity patterns. If reduced activity is the only source of phrase-level opacity and spreading is the only source of reduced activity, then only phonological processes that involve spreading should interact opaquely. Opacity is thus tied to the structural properties of a process. This excludes e.g. an opaque counterfeeding pattern between epenthesis and deletion. Since no spreading would be involved, the derivational history of the epenthetic and deleted segments could not be referred to by Shared Activity.

As far as the typology of opaque interactions is concerned, Shared Activation can de-

scribe both cases of underapplication (like in Kere) and overapplication/misapplication. Underapplication results if constraints on non-derived structures are less violated by multiple association and application to singly linked objects is preferred. Overapplication would result if constraints on derived structures are less violated by multiple association and application to singly linked objects is blocked, e.g. in Tiriki (Bantu, Kenya), where phrase-level downstep between adjacent high tones is blocked when tones become adjacent through phrase-level spreading (Paster & Kim, 2011). This constitutes a phonologically derived environment effect (Kiparsky, 1993), as downstep only applies if the context is derived by spreading. It can be analyzed by assuming a gradient constraint against TBUs with downstepped high tones that is fatally violated for singly-linked tones and violated to a non-fatal degree for multiply linked tones. Shared Activity is comparable and empirically distinguishable from other approaches to phrase-level opacity such as Super-Optimality (Kiparsky, 2011, 2016), Containment Theory (Zimmermann & Trommer, 2024), and theories with extrinsic rule ordering (Chomsky & Halle, 1968), as summed up in (15).

Super-Optimality is an anti-neutralization constraint introduced into Stratal Optimality Theory by Kiparsky (2011, 2016) in order to derive opaque counterfeeding in chain shifts. Super-Optimality blocks processes if the output could result from a different input, which would also provide a anti-neutralization approach to the Kere data. Any two processes could in principle stand in such a counterfeeding relation at the phrase-level, if they would otherwise result in a neutralization. Super-Optimality cannot derive phrasal counterbleeding patterns, but it unclear if it can derive phonologically-derived environment effects (Paul Kiparsky, personal communication).

Stratal Containment (Trommer, 2011) includes a version of constraint cloning, where constraints can either refer to the surface structure or to an integrated structure that includes both the complete surface structure and the complete underlying structure.

In testing the predictions of Containment Theory, Zimmermann & Trommer (2024) find that — for simple patterns — epenthesis can never counterbleed or counterfeed any process. Stratal Containment would thus predict that these patterns should not exist at the phrase-level. Theories with ordered rules at the phrase-level (e.g. Chomsky & Halle, 1968) can derive any counterfeeding and counterbleeding. Yet it provides no inherent solution to derived environment effects, similar to Containment Theory. Future research will have to show which predictions fit best with attested data.

(15) Partial comparison of predictions on phrase-level opacity

	Super-O.	Containm.	Shared Act.	Ord. R
Epenthesis counterfeeds spreading	✓	✗	✗	✓
Deletion counterfeeds spreading	✓	✓	✓	✓
Deletion counterfeeds epenthesis	✓	✓	✗	✓
Counterbleeding	✗	(✓)	(✗)	✓
Derived Environment Effects	?	(✗)	(✓)	(✗)

6 Conclusion

In this squib, I have presented a challenge for the hypothesis that there is no opacity in phrase-level phonology by introducing data from phrasal tone in Kere. Boundary Tone shift (and Low Deletion) provide a context for Low Deletion and Upstep, but neither process applies. This is problematic for several theories of phonology, which entail the No Phrase-Level Opacity Hypothesis. I have argued for modifying the representations to include gradient activity shared by TBUs associated to the same tone, which predicts underapplication by reducing the markedness violation of spread tones. This mechanism can be extended to non-tonal processes and makes the restrictive prediction that phrase-level opacity should always involve at least one spreading process.

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1. See, however, Baković (2007) for potential counterexamples and further discussion.
2. The example *yòblág-ná íglà* ‘inside our bones’ and the example *màré dímiyà* ‘to be near’ show an additional phrase-level rightward high tone spreading process, which seems to counterfeed Upstep.
3. In many languages, this process is known as plateauing (Cahill, 2007; Marlo & Odden, 2019).
4. SR only gives *gòlé* as the underlying form of *gólé* ‘old’ without surface forms in other contexts.
5. Boundary Tone Shift counterfeeds Low Deletion on environment and counterfeeds Upstep on focus; Low Deletion counterfeeds Upstep on focus (McCarthy, 1999).
6. Identical morphosyntax prevents a phrasal cyclicity account (cf. Clark, 1990; Sande et al., 2020).
7. The overline here indicates the repeating number sequence of a repeating decimal, i.e. $8.\overline{3} = 8.33\dots$
8. Most of the analysis would also work within a privative system of H, $\emptyset$, $^{\uparrow}$ H, except that one would need to modify the analysis of the Low Boundary Tone (see Jones (2014) for a similar reasoning).
9. The formulation could be changed to refer to a low-toned TBU between two TBUs “linked to high tones” instead of “linked to different high tones” if the constraint NOCROSSING is assumed to be either violable (cf. Bagemihl, 1989) and very high weighted or inviolable (Pulleyblank, 1996).
10. I assume that a general constraint against deleting tones MAX(Tone) is weighted low enough to allow for deletion of low tones in order to remedy violations of high-ranked markedness violations and a general DEP(T) constraint is weighted high enough to exclude this possibility for most contexts, except to satisfy the very high ranked * H% constraint.
11. See McCarthy (2002) and Trommer (2011) for constraints ignoring parts of the phonological structure.
12. NOSPREADR is only violable in Kere in order to provide tonal specifications for underspecified TBUs.
13. In (12), I take the activity of TBUs with contour tones to be the sum of the activity assigned for each of the tones, i.e. $a(\theta) = \sum_{k=1}^m \frac{a(\tau_k)}{n_k}$. Nothing hinges on this assumption and the assignment of activity for contour tones could in principle be different, e.g. by using the arithmetic mean instead.
14. The effects are similar to Head-Faith in Headed Spans (McCarthy, 2004), basic alignment in Optimal Domains (Cole & Kisseberth, 1994), or base faithfulness in reduplication (Zimmermann, 2021).
15. As a reviewer notes, this means that the account can derive unattested n-ary bounded vowel harmony. Since this is attested for small n for tones (cf. e.g. Breteler, 2017), the absence of n-ary spreading might be a learning issue instead, as argued by Stanton (2016) for the so-called midpoint pathology.
16. Smolensky et al. (2014) speculate that gradient activation could result in incomplete neutralization.